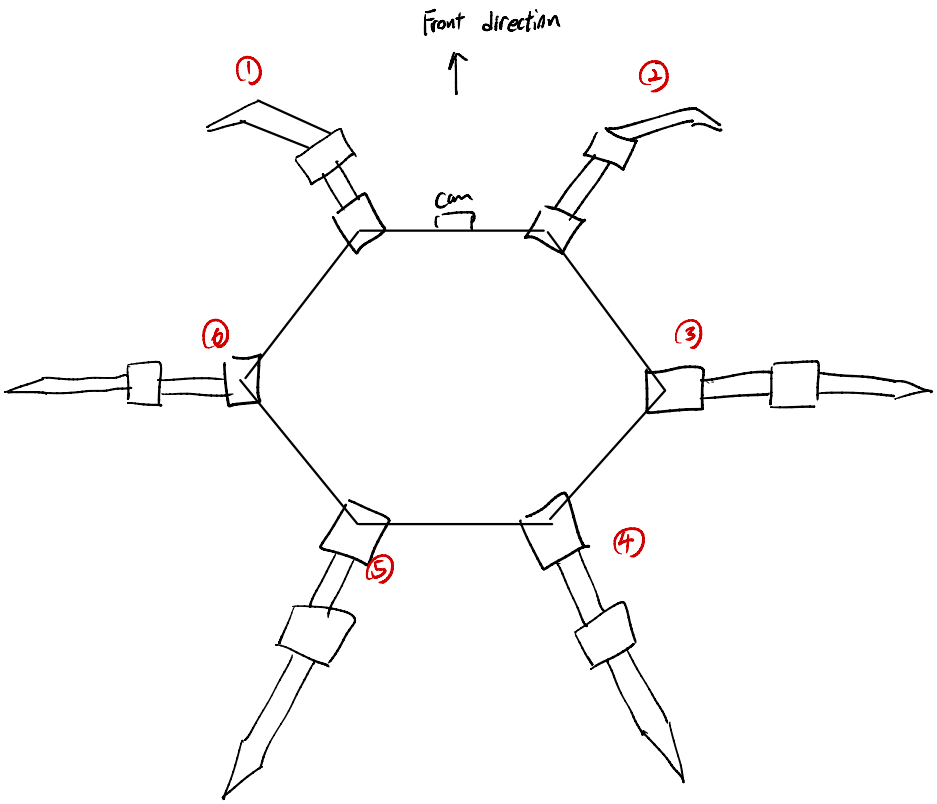
Draft :



Used Element :

Control Brain :

① ESP32 V1

② PCA 9685 PWM Driver x2

Power Management :

① 2S LiPo battery (7.4V)

③ Voltage Regular (Buck Convertor) :

DC to DC converter (down to 6V)

(High current 15A - 20A adjustable buck module)

③ Capacitors : Large Capacitor (e.g. 1000μF)

Body :

Leg Design : 3 DOF per leg :

①. Coxa : Forward / backward lifting

②. Femur : Leg lifting

③. Tibia : Extending / retracting leg.

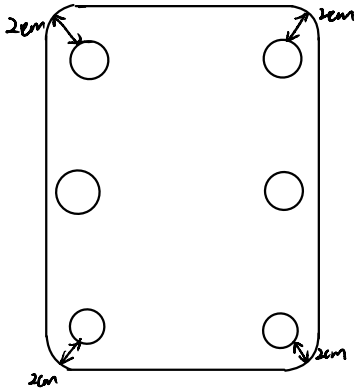
Material : 3D printed / laser-cut acrylic / aluminium.

Software & Kinematic :

① Inverse Kinematic

② Gait Algorithms.

Body :



single leg :

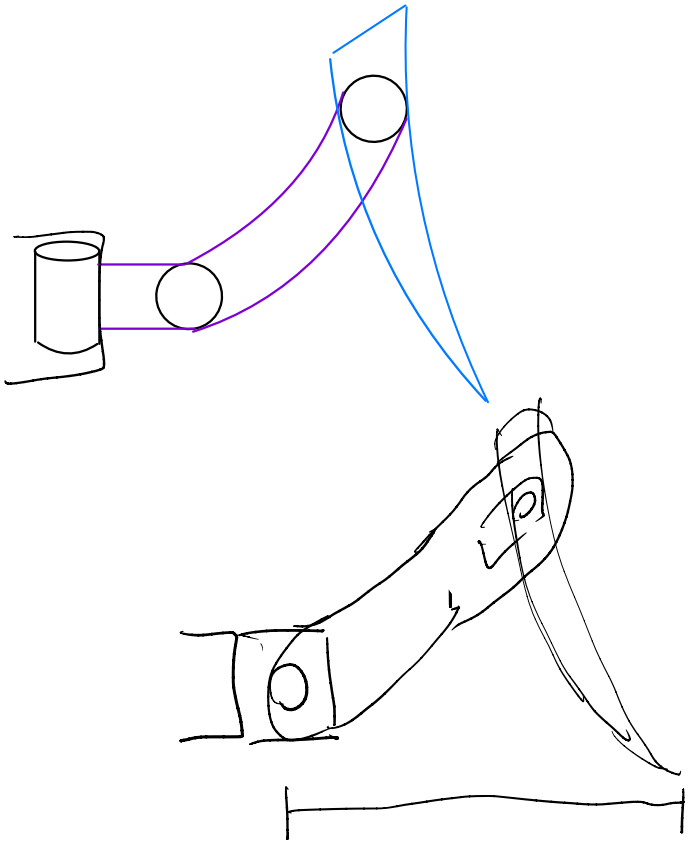
3 x MG90S servo motor

body :

1 x ESP32

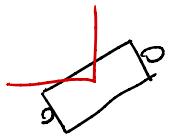
2 x PCA9685

Single leg

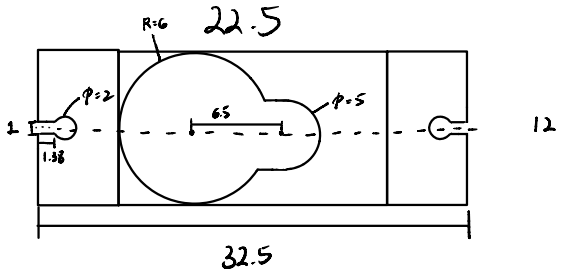


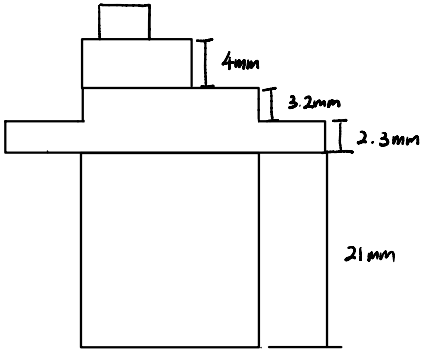
< 435

Coxa :



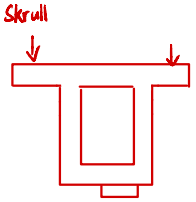
MG90S :

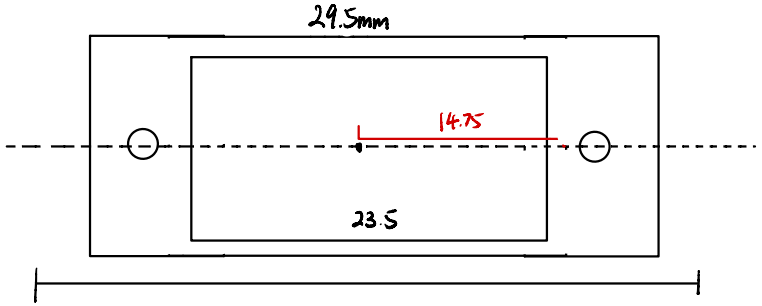




Coxa Mount : Wall : 3mm

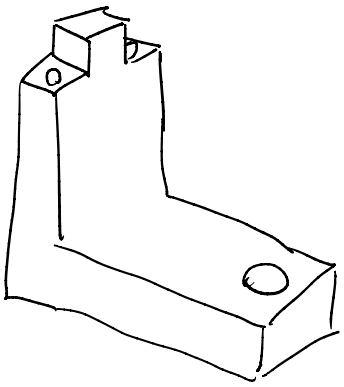
tolerance : 0.5mm each side





Femur Mount :

bottom :



upper :

